Robust Dedicated Actuator Placement via Matroid Spanning

Qichao Wang
Hebei University of Technology

Daoyu Dong
University of Electronic Science and Technology of China

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Abstract

We study dedicated actuators installed at distinct states of a structured linear system, where failures remove actuators but leave all states and state edges intact. For unit costs and one failure, an exact source-quota relaxation followed by maximum matching in a common-repair graph gives a 3/2-approximation on arbitrary directed state graphs in $O(n^3)$ graph operations. The proof applies the Tutte–Berge formula to disjoint unions of fundamental cocircuits. For multiple failures, we prove exact fixed-parameter tractability in the maximum-matching deficiency plus the failure budget, independently of the number of source strongly connected components. The underlying theorem optimizes robust matroid spanning with arbitrarily many disjoint unit-cost quotas. We also prove a weighted safe/vulnerable spanning theorem in the independence-oracle model, with a control application on strongly connected graphs.

1 Introduction

A structurally controllable system can lose controllability when an actuator fails. Redundant placement must protect both the matching condition and accessibility from the installed inputs. These requirements have different combinatorial forms: robust spanning in a matroid and lower quotas on source strongly connected components. Keeping both requirements explicit permits approximation and exact parameterized algorithms within the same model.

Our first result is a cubic-time 3/2-approximation for one actuator failure and unit costs (Theorem 3.3). Its repair argument is valid for arbitrary matroids. The exact graph relaxation, not a minimum ordinary controllability placement, provides the necessary lower bound. A three-state example attains the ratio for a permissible relaxation choice.

Our second result concerns arbitrary independence-oracle matroids with nonnegative purchase weights and safe or vulnerable elements (Theorem 4.1). Building on the uniform-set core method of Bentert et al. [5], a weight-ordered recursion preserves an optimum robust purchased set. The recursion continues into lower-rank closures even when a large uniform set has been found. A classical set-pairs inequality sharpens the deficient-hyperplane count. The parameter is rank plus failure budget, and the objective charges every purchased element.

The third result is exact unit-cost optimization with arbitrarily many disjoint quota groups (Theorem 5.1). A bounded robust core need not contain all elements needed for accessibility. We instead preserve its quota-completion cost by capping the number of representatives of each group. The resulting control algorithm is FPT in matching deficiency plus failure budget on arbitrary digraphs (Corollary 5.2); the number of source components is not a parameter.

Poças, Pequito and Ramos [13] study the robust minimal structural controllability problem, establishing NP-hardness and reporting approximation factors 5/2 for general digraphs and 2 for strongly connected digraphs. We specify the actuator-failure conventions in Section 2 and discuss related robust-assignment and fault-tolerant matroid algorithms in Section 6.

2 Model and the dual matching matroid

Let $G = (E, \mathcal{A})$ be a nonempty finite directed state graph with $n = |E|$; self-loops are allowed. Each prescribed nonzero entry of the state and input matrices is a separate free parameter. Structural controllability means that a controllable numerical realization exists, equivalently that almost every realization is controllable. A placement $P \subseteq E$ installs one dedicated input at each selected state, at most one per state. An actuator failure removes the installed input while the state vertices and arcs remain intact; the placement is fixed before failures occur.

Write $S_1, \dots, S_s$ for the source strongly connected components (SCCs), those having no entering state edge from another SCC. Form the state bipartite graph with a left and right copy of E, and an edge $u_L v_R$ for each arc $u \to v$. Let T be the transversal matroid on the right copies: its independent sets are precisely the state sets matchable using state edges. Put $M = T^*$ and $d = n - \operatorname{rank}_T(E)$, the maximum-matching deficiency. For $X \subseteq E$, duality gives

$$\operatorname{rank}_M(X) = |X| + \operatorname{rank}_T(E \setminus X) - \operatorname{rank}_T(E). \tag{1}$$

In particular, X spans M if and only if $E \setminus X$ is independent in T. A rank oracle for M is computable by bipartite matching.

The standard structural-controllability criterion combines accessibility with a matching saturating all state vertices [9, 12]. With dedicated inputs it yields the following useful formulation.

Proposition 2.1. *A placement P tolerates every set of at most k actuator failures if and only if*

$$\operatorname{rank}_M(P \setminus F) = d \quad (F \subseteq P,\ |F| \le k), \qquad |P \cap S_i| \ge k + 1 \quad (1 \le i \le s). \tag{2}$$

Proof. For a fixed surviving placement X, matching every state in $E \setminus X$ by state edges and every state in X by its own input gives a matching saturating all states. Conversely, an input-augmented saturating matching matches every state in $E \setminus X$ using state edges. This proves the matching equivalence, and (1) gives spanning in M. Accessibility holds exactly when every source SCC contains an input: every component is reachable from a source component in the condensation DAG, and no input outside a source component can reach it. The source SCCs retain an input after every at most k failures exactly when they initially contain at least $k + 1$. Applying both criteria to every survivor proves the claim. □

We use standard finite-matroid notation. The closure of X is $\operatorname{cl}(X) = \{e : \operatorname{rank}(X \cup \{e\}) = \operatorname{rank}(X)\}$. A hyperplane is a flat of rank one below the matroid rank. A loop has rank zero; a coloop belongs to every basis. The coloops of a restriction $M|P$ are precisely the elements whose deletion lowers $\operatorname{rank}_M(P)$. A circuit is a minimal dependent set. For a basis B and $e \notin B$, the unique circuit contained in $B \cup \{e\}$ is its fundamental circuit [11].

3 A Cubic-Time 3/2-Approximation

Throughout this section $k = 1$ and every actuator has unit cost.

3.1 An Exact Source-Quota Relaxation

Minimize $|P|$ subject to P spanning M and $|P \cap S_i| \geq 2$ for all i. Denote its optimum by L. Every robust placement is feasible for this relaxation, so $L \leq \mathrm{OPT}$.

It is solvable by an integral maximum flow. The complement $I = E \setminus P$ must be matchable using state edges and must satisfy $|I \cap S_i| \leq |S_i| - 2$. Use a source, the two state copies, one group node per source SCC, and a sink. Source-to-left-state edges have capacity one; left-to-right state edges have capacity one. Each right state has a capacity-one outgoing edge, to its source group if it has one, and directly to the sink otherwise. The edge from group i to the sink has capacity $|S_i| - 2$. An integral maximum flow gives a maximum-cardinality admissible I, and therefore a minimum-cardinality P.

A singleton source SCC makes robust placement impossible. Conversely, if all source SCCs have at least two vertices, every vertex has an incoming state edge: this is immediate within a nonsingleton SCC, and a singleton nonsource SCC has an entering edge. Thus T has no loops and M has no coloops. The full placement E spans after every single deletion and satisfies all quotas. This proves that singleton source SCCs characterize infeasibility for this one-failure model.

3.2 Cocircuit unions and a global lower bound

The next two lemmas concern an arbitrary rank-d matroid M with a spanning set P. Let D be the coloops of $M|P$. For $b \in D$ define

$$C_b = E \setminus \mathrm{cl}(P \setminus \{b\}).$$

The set C_b restores spanning after deleting b and meets P exactly in $\{b\}$.

Lemma 3.1. *For every $R \subseteq D$,*

$$\mathrm{rank}\Big(E \setminus \bigcup_{b \in R} C_b\Big) = d - |R|.$$

Proof. Choose a basis $B \subseteq P$. Every $b \in D$ belongs to B, and $B \setminus \{b\} \subseteq P \setminus \{b\}$ are both of rank $d - 1$. Their closures are consequently equal. For nonempty R we claim

$$\bigcap_{b \in R} \mathrm{cl}(B \setminus \{b\}) = \mathrm{cl}(B \setminus R).$$

For $e \notin B$, membership in $\mathrm{cl}(B \setminus \{b\})$ means that its fundamental circuit omits b. Membership for every $b \in R$ therefore means that this circuit is contained in $(B \setminus R) \cup \{e\}$, which is precisely membership in $\mathrm{cl}(B \setminus R)$. Basis elements satisfy the equality directly, and taking ranks proves the identity. For $R = \varnothing$ it is simply $\mathrm{rank}(E) = d$. □

Form a simple undirected, possibly nonbipartite graph K on D, with an edge bc when some $e \in E \setminus P$ belongs to both C_b and C_c. Retain one such witness for each edge, and let ν denote the maximum matching cardinality of K.

Lemma 3.2. *Every set W spanning M after any one deletion satisfies $|W| \geq 2(|D| - \nu)$.*

Proof. Fix $U \subseteq D$ and let $R_1, \ldots, R_t$ be the nonempty connected components of $K - U$. Put $Q_i = \bigcup_{b \in R_i} C_b$. These sets are pairwise disjoint. An element of P belongs to at most one C_b. A common external element of Q_i, Q_j would witness an edge between their components. Lemma 3.1 gives $\mathrm{rank}(E \setminus Q_i) = d - |R_i|$. Since adding an element raises rank by at most one, W must contain at least $|R_i|$ elements of Q_i to span. Equality is impossible: deleting one of these elements would leave rank at most $(d - |R_i|) + (|R_i| - 1) < d$. Thus

$$|W| \geq \sum_i (|R_i| + 1) = |D| - |U| + t \geq |D| - |U| + \mathrm{odd}(K - U),$$

where odd counts odd-order components. The classical Tutte–Berge formula [10] states

$$2\nu = |D| - \max_{U \subseteq D} \big(\mathrm{odd}(K - U) - |U|\big).$$

Taking the maximum in the preceding inequality proves the result. □

3.3 Algorithm and guarantee

Theorem 3.3. *For any nonempty directed state graph in the stated dedicated-actuator model, a robust placement of size at most* $3\mathrm{OPT}/2$ *can be found, or infeasibility reported, in* $O(n^3)$ *graph operations and* $O(n^2)$ *space. The guarantee is for unit costs and one actuator failure.*

Proof. Reject singleton source SCCs; otherwise compute an optimum relaxation P. Find D and a maximum matching of K. For each matched pair install its external common-repair witness. For every unmatched $b \in D$, install any external element of C_b. Such an element exists because M has no coloops, while $P \setminus \{b\}$ has rank $d - 1$. Let R be the set of installed witnesses, with repetitions removed. There are at most $\nu + (|D| - 2\nu) = |D| - \nu$ witnesses.

The set $P \cup R$ is robust. If a witness fails, P still spans. If an original noncritical element fails, the surviving part of P spans. If $b \in D$ fails, its witness repairs the deficiency. The source quotas persist under additions. By Lemma 3.2,

$$|P \cup R| \le L + |D| - \nu \le \mathrm{OPT} + \frac{\mathrm{OPT}}{2}.$$

For the complexity, retain the state matching J obtained from the quota flow. It covers $I = E \setminus P$. For each $b \in P$, explore alternating paths from unmatched right vertex b: traverse state edges to the left, then matched edges back to the right. A reachable free left vertex is exactly an augmenting path showing that $I \cup \{b\}$ is matchable, or that b is noncritical. For a critical b, the reachable matched right vertices are exactly its external repairs. A path to right vertex e can be flipped to match $I \setminus \{e\} \cup \{b\}$. Conversely, the symmetric difference with any matching of that replacement set gives such a path, since a path to a free left vertex would contradict criticality. Hence these searches recover the complete repair supports.

There are $O(n)$ vertices and $O(n^2)$ edges in the flow network. At most n unit augmentations, each found in $O(n^2)$ operations, suffice. At most n alternating searches also cost $O(n^3)$. For each external state, connect all pairs of critical states it repairs, storing a witness for each edge. This clique construction costs $O(n^3)$. A maximum general-graph matching is computable in $O(n^3)$ operations by a blossom algorithm [10]; a maximal matching is insufficient. All structures require $O(n^2)$ space. □

For the bidirected three-state star with center a and leaves b, c, the relaxation can choose $P = \{a, c\}$. Its one critical state is c, and the only external repair is b. The algorithm returns all three states, while $\{b, c\}$ is robust. Thus the ratio $3/2$ is attained for this relaxation choice. Choosing the optimum robust pair at the relaxation stage avoids this worst case.

4 Weighted Robust Matroid Spanning

Let $M = (E, \mathcal{I})$ be a rank-r independence-oracle matroid, $V \subseteq E$ the vulnerable elements, $E \setminus V$ the safe elements, and $k \ge 0$ the failure budget. A set B is *resilient* if

$$\mathrm{rank}_M(B \setminus F) = r \quad \text{for every } F \subseteq B \cap V, \quad |F| \le k. \tag{3}$$

For nonnegative binary-encoded weights w, the objective is to minimize $w(B) = \sum_{e \in B} w(e)$.

Theorem 4.1. *An optimum resilient purchased set can be computed, or infeasibility reported, in $f(r,k)\,\mathrm{poly}(n,\ell)$ time and oracle calls, where ℓ is the input bit length and independence queries have unit cost. For $r \geq 2$, writing $q = (k+1)r$, a bound is*

$$2^{O(qr^2(r+k))}\,\mathrm{poly}(n,\ell),$$

hence $2^{O((r+k)^5)}\,\mathrm{poly}(n,\ell)$. The cases $r \leq 1$ or $k = 0$ are polynomial-time solvable.

4.1 Matroid preliminaries and a small optimum

Loops can be discarded because their removal preserves resilience and cannot increase nonnegative purchase cost. If $r = 0$, the empty set is optimal. If $k = 0$, a minimum-weight basis is optimal by matroid greedy. For $r = 1$, choose the cheaper feasible option between a cheapest safe nonloop and the $k+1$ cheapest vulnerable nonloops. Henceforth assume $r \geq 2$ and $k \geq 1$, and work on the loop-free restriction. Set

$$q = (k+1)r, \qquad H = \binom{r+k-1}{k}, \qquad t_h = (h-1)H + q \quad (1 \leq h \leq r). \tag{4}$$

Lemma 4.2. *Every resilient set B contains a resilient subset B_0 of cardinality at most q. Consequently, if a feasible solution exists, there is a minimum-weight solution of cardinality at most q.*

Proof. Successively choose a maximal independent subset I_j of $B \setminus \bigcup_{i<j} I_i$, for $j = 0, \ldots, k$, allowing empty layers, and put $B_0 = \bigcup_{j=0}^{k} I_j$. Since $|I_j| \leq r$, we have $|B_0| \leq q$.

Fix $F \subseteq B_0 \cap V$ with $|F| \leq k$. The layers are disjoint, so some I_j avoids F. Every element of $B \setminus B_0$ was available when I_j was chosen and belongs to $\mathrm{cl}(I_j)$. Every element of $B_0 \setminus F$ already lies in that surviving set. Therefore

$$B \setminus F \subseteq \mathrm{cl}(B_0 \setminus F).$$

Since F is also an allowed failure of B, its left-hand side spans M. Thus B_0 is resilient. Nonnegative weights give $w(B_0) \leq w(B)$, and applying this construction to an optimum proves the second assertion. □

4.2 A Weight-Respecting Exchange

The classical set-pairs inequality of Bollobás [6] yields the following sharp deficient-hyperplane bound.

Lemma 4.3. *For any $Y \subseteq E$, at most $\binom{r+k-1}{k}$ distinct hyperplanes can have the form $\mathrm{cl}(Y \setminus F)$ with $F \subseteq Y$ and $|F| \leq k$. This count is sharp.*

Proof. Index these hyperplanes by P_i. Choose a basis $A_i \subseteq Y \cap P_i$ of size $r-1$, and put $D_i = Y \setminus P_i$. Because $P_i = \mathrm{cl}(Y \setminus F)$ for some permitted F, we have $D_i \subseteq F$, so $|D_i| \leq k$. Also $A_i \cap D_i = \varnothing$. If $i \neq j$, then $A_i \cap D_j \neq \varnothing$: otherwise $A_i \subseteq P_j$ and hence $P_i = \mathrm{cl}(A_i) \subseteq P_j$, forcing equality of the two hyperplanes.

In a uniformly random permutation of Y, let $\mathcal{E}_i$ be the event that every element of A_i precedes every element of D_i. Its probability is $\binom{r-1+|D_i|}{r-1}^{-1}$. These events are pairwise disjoint. Indeed, for $i \neq j$ choose $a \in A_i \cap D_j$ and $b \in A_j \cap D_i$. The two events require a before b and b before a, respectively. Therefore

$$\sum_i \binom{r-1+|D_i|}{r-1}^{-1} \leq 1.$$

Each summand is at least $\binom{r+k-1}{r-1}^{-1}$, proving the bound. For sharpness take $M = U_{r,r+k}$, $B = E$, all elements vulnerable, and $Y = B \setminus \{x\}$. The set B is resilient, and each k-subset of Y leaves a distinct $(r-1)$-element hyperplane. Thus equality holds even for the deficient hyperplanes needed below. □

Call Z *h-uniform* if $\operatorname{rank}(Z) = h$ and every h-element subset of Z is independent. Such a set has at least h elements, and every subset of size at most h is independent. A set has a single failure status if it lies entirely in V or entirely in $E \setminus V$.

Lemma 4.4. *Let $A \subseteq E$ have a single failure status and rank $h \geq 1$. Let $Z \subseteq A$ be h-uniform with $|Z| \geq t_h$. Suppose B is resilient, $|B| \leq q$, and $x \in (B \cap A) \setminus Z$ satisfies $w(z) \leq w(x)$ for every $z \in Z$. Then some $y \in Z \setminus B$ makes*

$$B' = (B \setminus \{x\}) \cup \{y\}$$

resilient, with $|B'| = |B|$ and $w(B') \leq w(B)$.

Proof. Put $Y = B \setminus \{x\}$. Consider the permitted failures $F \subseteq Y \cap V$, $|F| \leq k$, for which $\operatorname{rank}(Y \setminus F) < r$. Resilience of B implies that their surviving ranks equal $r-1$ and that $x \notin \operatorname{cl}(Y \setminus F)$. Lemma 4.3 bounds the number of these distinct hyperplanes by H. It applies also when x is safe and some deficient failure sets have size less than k.

Each of these hyperplanes contains at most $h-1$ elements of Z. Otherwise it contains an independent h-subset of Z, which spans A, and hence contains x, a contradiction. At most $q-1$ elements of Z belong to B, because $x \notin Z$. The number of forbidden choices in Z is at most $(h-1)H + q - 1 < t_h$. Choose a $y \in Z \setminus B$ outside all the deficient hyperplanes.

Let $F' \subseteq B' \cap V$ have size at most k. If $y \notin F'$, then $F' \subseteq Y$; either $Y \setminus F'$ already spans or y repairs its rank deficiency by construction. If $y \in F'$, then y is vulnerable, so x is vulnerable as well. The set

$$(F' \setminus \{y\}) \cup \{x\}$$

is an allowed failure of B of the same size as F'. Its survivor is exactly $B' \setminus F'$, which therefore spans. The cardinality and weight assertions follow from the choice of y. □

4.3 Recursive Construction of an Optimum-Preserving Core

For a loop-free set A of a single failure status, define $\operatorname{Core}(A)$ by the following procedure. Every scan uses a fixed total ordering that refines nondecreasing weight.

1. If $A = \varnothing$, return $\varnothing$. Set $h = \operatorname{rank}(A)$. If $|A| \leq t_h$, return A.

2. Scan the elements of A in order, initially with $Z = \varnothing$. Accept an element x precisely when every subset of $Z \cup \{x\}$ of size at most h is independent. Stop when $|Z| = t_h$ or all elements have been scanned.

3. If $h = 1$, return Z. Otherwise, for every $T \in \binom{Z}{h-1}$, put $A_T = A \cap \operatorname{cl}(T)$ and recursively compute $\operatorname{Core}(A_T)$. Return

$$Z \cup \bigcup_{T \in \binom{Z}{h-1}} \operatorname{Core}(A_T).$$

The recursive step applies whether the scan reaches its threshold or exhausts A. The parameters q, H remain fixed across recursive calls.

Until Z has size h, the acceptance rule is ordinary greedy independence. The final Z therefore has at least h elements and is h-uniform. Each child has rank exactly $h-1$: it contains T and is contained in $\operatorname{cl}(T)$.

Lemma 4.5. *For every resilient B with $|B| \le q$ and every single-status loop-free set A, there is a resilient B' such that*

$$|B'| = |B|, \quad w(B') \le w(B), \quad B' \setminus A = B \setminus A, \quad B' \cap A \subseteq \mathrm{Core}(A).$$

Proof. Induct on $h = \mathrm{rank}(A)$. Empty domains and a return of all of A require no change. For $h = 1$, every scanned nonloop is accepted, so the threshold scan retains the q lightest elements. Lemma 4.4 replaces every selected element outside Z at no greater weight, proving the base case.

Suppose $h > 1$ and the scan is executed. Let $U = \bigcup_T A_T$. Every rejected scanned element x belongs to U. Indeed, at rejection a dependent subset of $Z \cup \{x\}$ contains x and at most $h - 1$ retained elements. Those retained elements are independent, so they span x. Extend them to an $(h-1)$-subset of the final Z, obtaining a child containing x. Also $Z \subseteq U$, since $h > 1$ and every element of Z belongs to an $(h-1)$-subset of Z.

If the scan is exhausted, $A = U$. Otherwise each element of $A \setminus U$ is unscanned and is at least as expensive as every element of Z. Apply Lemma 4.4 successively to the elements of $B \cap (A \setminus U)$. Each replacement lies in $Z \subseteq U$ and removes one selected element outside U. The resulting resilient set has its intersection with A inside U, unchanged cardinality and outside-A part, and no larger weight.

Fix the final output $W_A = \mathrm{Core}(A)$ and process the children $A_1, \dots, A_m$ in any order. Apply the induction hypothesis to each child and the current solution. For the current solution define $D(B) = (B \cap A) \setminus W_A$. Processing A_i changes only elements in A_i and leaves its selected elements in $\mathrm{Core}(A_i) \subseteq W_A$. Consequently,

$$D(B_{\mathrm{new}}) \subseteq D(B_{\mathrm{old}}) \setminus A_i.$$

Initially $D(B) \subseteq U$. After all children have been processed it is empty, regardless of overlaps between children. All operations preserve cardinality, hence the bound q, and preserve resilience without increasing weight. This establishes the induction. □

Apply the construction separately to the vulnerable and safe nonloop elements, and let W be the union of the two outputs. Their domains are disjoint. Lemmas 4.2 and 4.5 imply that W contains an optimum of cardinality at most q whenever the instance is feasible.

4.4 Size and running time

Put $K_0 = 0$, $K_1 = q$, and, for $2 \le h \le r$, define

$$K_h = t_h + \binom{t_h}{h-1} K_{h-1}.$$

By induction, a rank-h call returns at most K_h elements. The sequence is nondecreasing; in particular $|W| \le 2K_r$, or $|W| \le K_r$ if all elements are vulnerable.

For an explicit bound put $\tau = t_r$ and $N = 2r\tau^{(r-1)^2}$. Each of the two recursion trees has at most r levels and branching at most τ^{r-1}, giving at most N nodes and $|W| \le N\tau$. Each node uses $O(n(1 + \tau^{r-1}))$ independence queries for the uniformity scan and $O(n\tau^{r-1})$ queries for its lower-rank closures. Rank computation, sorting, and bookkeeping incur polynomial input cost times a parameter-only factor.

Finally enumerate all $B \subseteq W$ with $|B| \le q$, and for each enumerate all allowed failure sets and compute its surviving ranks greedily. There are at most $(N\tau + 1)^q$ candidates and at most 2^q failures per candidate. Choose a feasible candidate of minimum total weight. If there is none, optimum preservation proves infeasibility. Exact weight sums and comparisons have polynomial bit cost in the original weight encoding.

For $r \geq 2$, put $p = r + k$. Since $H \leq 2^{p-1}$ and $q \leq (p+1)^2$, we have $\log_2 \tau = O(p)$. Hence $N\tau = 2^{O(r^2 p)}$, and enumerating the candidates and failures gives $2^{O(qr^2 p)} \operatorname{poly}(n, \ell)$. As $q = O(p^2)$, this is at most $2^{O(p^5)} \operatorname{poly}(n, \ell)$. For fixed $k \geq 1$, the sharper estimate $H = O_k(r^k)$ gives $\tau = O_k(r^{k+1})$, $N\tau = r^{O_k(r^2)}$, and running time $r^{O_k(r^3)} \operatorname{poly}(n, \ell)$.

4.5 Why simpler weighted reductions do not suffice

Stopping at a large uniform set. Consider a rank-two truncation of a partition matroid with two parallel classes of size two having weights $(0, 0)$ and $(1, 1)$, and five further singleton classes each of weight 100. Let $k = 1$. Then $q = 4$, $H = 2$, and $t_2 = 6$. The weight-ordered uniform scan reaches the threshold while retaining only one element from each cheap parallel class. Its robust optimum is 101, whereas the full instance has optimum 2, attained by the two cheap pairs. Thus sorting the scan is insufficient if a successful threshold causes the recursion to stop. Our construction retains the cheap rejected elements through the lower-rank children.

Preserving cheapest ordinary bases after failures. Dey and Kavitha [7] study a preserver $S \subseteq E$ such that, for every $F \subseteq E$ with $|F| \leq k$, $M|(S \setminus F)$ contains a minimum-cost ordinary basis of $M|(E \setminus F)$. That property need not preserve the optimum total cost of a resilient purchased set.

For a five-element example, take a rank-two truncation with parallel classes A, B of size two and a singleton class C. Give each element of $A \cup B$ weight 2 and the element of C weight 3. For every zero or one ground-element failure, $A \cup B$ still contains an ordinary basis of cost 4, which is minimum in the full surviving matroid. Yet the robust optimum inside $A \cup B$ is 8, and the full instance has optimum 7, using one element from each class. Two successive cheapest residual bases also produce $A \cup B$. Thus this preserver construction does not replace the optimum-preserving recursion above.

4.6 An application to dedicated input placement

Consider a nonempty strongly connected state digraph with nonnegative actuator costs and specified safe or vulnerable locations. Keep the single-actuator-per-state and actuator-only failure conventions of Section 2. Its dual state matching matroid has rank equal to the deficiency d; Proposition 2.1 supplies the matching interpretation, also when only vulnerable actuators may fail.

Corollary 4.6. *Minimum-cost robust dedicated-input placement on a strongly connected state digraph, under the actuator-failure model just specified, is FPT in $d + k$, with nonnegative binary-encoded costs and safe/vulnerable actuator locations.*

Proof. If $d > 0$, every surviving spanning set in M is nonempty. Strong connectivity then ensures accessibility, so Theorem 4.1 applies. If $d = 0$, matching is automatic and accessibility requires a nonempty survivor. Choose the cheaper of one cheapest safe actuator and $k + 1$ cheapest vulnerable actuators, omitting unavailable options. □

For general state digraphs, additional source SCC constraints are necessary. They are not preserved by the weighted same-status exchange alone. Section 5 supplies a separate unit-cost quota theorem, not a general weighted extension.

5 Exact Parameterized Optimization with Source Quotas

On a general state digraph, spanning must be combined with source-SCC accessibility quotas. The following matroid theorem permits an arbitrary number of disjoint quota groups, with unit costs and all elements vulnerable.

Theorem 5.1. *Let M be a rank-r matroid on E, given by an independence oracle, and let $S_1, \dots, S_s \subseteq E$ be pairwise disjoint. For $k \geq 0$, a minimum-cardinality set B satisfying*

$$\operatorname{rank}_M(B \setminus F) = r \quad (F \subseteq B,\ |F| \leq k), \qquad |B \cap S_i| \geq k+1 \quad (1 \leq i \leq s)$$

can be computed, or infeasibility certified, in $f(r,k)\,\mathrm{poly}(|E|)$ oracle and arithmetic operations.

Proof. Set $\beta = k+1$, $q = \beta r$, and $G_0 = E \setminus \bigcup_i S_i$. A group with fewer than β elements makes the instance infeasible. For $r = 0$, the optimum consists of β elements from each group. Assume $r \geq 1$.

By Lemma 4.2, a resilient spanning set has a resilient subset C of size at most q, which can be chosen without loops; loops remain available in the ground set for quota completion. Define

$$\phi(C) = |C \cap G_0| + \sum_{i=1}^{s} \max\{0, |C \cap S_i| - \beta\}.$$

Completing the quotas has cost $\beta s + \phi(C)$, so minimizing ϕ over resilient cores of size at most q yields an optimum. The completion cost must be preserved: in $U_{1,4}$ with $k = 1$ and $S_1 = \{c, d\}$, the core $\{a, b\}$ completes to four elements, while $\{c, d\}$ is already optimal.

Color-capped core. Set $H = \binom{r+k-1}{k}$ and

$$c_h = (h-1)H + q, \qquad T_h = (q+1)c_h.$$

Lemma 4.4, specialized to unit weights and all elements vulnerable, supplies a robust same-cardinality exchange from any h-uniform candidate set of size at least c_h. Start with the nonloops $E^\circ = \{e : \operatorname{rank}_M(\{e\}) = 1\}$, regarding $G_0, S_1, \dots, S_s$ as colors. On a rank-h domain A, scan into an h-uniform set Z with at most c_h accepted elements per color, stopping at $|Z| = T_h$ or exhaustion. Return Z if $h = 1$; otherwise, together with Z, retain the recursively computed cores of

$$A \cap \mathrm{cl}_M(Q), \qquad Q \in \tbinom{Z}{h-1}.$$

Before h elements have been accepted, the color cap cannot bind because $c_h \geq q \geq r \geq h$. Thus $\operatorname{rank}(Z) = h$ and every child has rank $h-1$.

Optimum preservation. We induct on h to prove that every resilient B with $|B| \leq q$ can be replaced by a resilient set of the same size, unchanged outside A, with A-part in the returned core and nonincreasing ϕ. Write $U = Z$ when $h = 1$ and otherwise let U be the union of the child domains. As in Lemma 4.5, dependence-rejected elements lie in U, and $Z \subseteq U$.

Consider $x \in (B \cap A) \setminus U$. If x was examined by the scan, it was rejected at its color cap. The c_h retained elements of that color admit a replacement by Lemma 4.4, preserving all color counts and hence ϕ. Otherwise the total threshold was reached. With $Y = B \setminus \{x\}$, discard the color G_0 and any group satisfying $|Y \cap S_i| \geq \beta$. At most q colors are discarded, since $|Y| \leq q-1$ and $\beta \geq 1$; they cover at most qc_h points of Z. At least c_h candidates remain in unsaturated groups, so Lemma 4.4 supplies an exchange that adds zero to ϕ, while removing x cannot increase it.

Repeated exchanges place the selected elements in U, finishing the rank-one case. At higher rank, apply the induction hypothesis to the child domains. The decreasing residual-set argument from the proof of Lemma 4.5 handles overlapping children, and each exchange preserves the cost inequality. This proves the invariant.

Running time. The size of the resulting core is bounded by L_r where

$$L_1 = T_1, \qquad L_h = T_h + \binom{T_h}{h-1} L_{h-1} \quad (2 \le h \le r).$$

This is a function of r, k; the recursion depth is at most r, and each node performs polynomially many oracle operations for fixed r, k. Enumerating all core subsets of size at most q, checking up to 2^q failures for each, minimizing ϕ, and completing the quotas gives the stated running time and an optimum placement. □

Corollary 5.2. *For arbitrary directed state graphs, minimum unit-cost dedicated-actuator placement tolerating at most k actuator failures is fixed-parameter tractable in the maximum-matching deficiency d plus k, with at most one actuator per state.*

Proof. Use the dual of the state matching transversal matroid, with rank $r = d$, the state maximum-matching deficiency. Surviving matching feasibility is spanning in this dual, and surviving accessibility is equivalent to selecting at least $k + 1$ states in each source SCC. The latter sets are disjoint, so Theorem 5.1 applies. □

6 Relationship to earlier work and limits

The robust purchased-set objective differs from choosing an ordinary basis after failures. Bentert et al. [5] prove unweighted FPT in rank plus failure budget and explicitly pose the weighted question. Their proceedings Proposition 6 gives the layered cardinality bound, Lemma 8 the uniform-set exchange and basis-subset hyperplane count, and Lemma 9 the recursive core. These are inherited ingredients. Our weighted argument adds the ordered exchange invariant, recursion in both stopping cases, and separation of failure statuses. The sharper count is an application of Bollobás's classical set-pairs inequality [6]. The same predecessor already gives polynomial weighted algorithms for rank at most two and for truncations of unit-capacity partition matroids.

The all-vulnerable case of Theorem 4.1 addresses the weighted problem in Section 1.1 of [5]. In personal correspondence with Qichao Wang (5 November 2026), Matthias Bentert confirmed that the condition $\operatorname{rank}(M - F) = \operatorname{rank}(M)$ printed in its Section 7 was intended as $\operatorname{rank}_M(B \setminus F) = \operatorname{rank}(M)$, the selected-set formulation used here and in related fault-tolerant design [3].

Adjiashvili, Bindewald and Michaels [1], Theorem 7, give a 7/4-approximation for uniform cardinality node-robust assignment. Their repair perspective is a predecessor of the present approach. Their ICALP 2016 Theorem 5 [2] already gives 3/2 for uniform *edge*-robust assignment, buying assignment edges and tolerating an edge failure via ear decompositions. That is a different purchase/failure model. The proof in Section 3 instead uses fundamental cocircuit unions, maximum common-repair matching, and source quotas.

Bulk-robust optimization [4] charges purchased elements but supplies an explicit family of failure scenarios. Expanding all at most k failures can require $n^{\Theta(k)}$ scenarios, so polynomial time in that expanded list does not establish the present FPT bound. Weighted representative families [8] preserve independent extensions under their stated representation and matroid hypotheses; their predicate is not directly optimum preservation for a dependent purchased set under all failures. Likewise, the cheapest-surviving-basis preservers of Dey and Kavitha [7] optimize a different object, as the example above shows.

The weighted theorem applies to safe and vulnerable elements, whereas the quota theorem assumes unit costs and that all selected elements may fail. Extending the latter to weighted quotas is an open direction. The optimal polynomial-time approximation factor also remains open: the three-state equality example concerns our relaxation choice, not an inapproximability bound for the problem.

Code availability. Implementations and verification scripts are available at `https://github.com/dongxuelian2/robust-structural-controllability`. The approximation and parameterized implementations are linked to fixed source revisions.

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